Journal of Monetary Economics 155 (2025) 103823

Contents lists available at ScienceDirect

Journal of Monetary Economics

journal homepage: www.elsevier.com/locate/jme

Hedge funds and the Treasury cash-futures basis trade ☆

Daniel Barth, R. Jay Kahn *

Board of Governors of the Federal Reserve System, 20th Street and Constitution Avenue N.W., Washington, DC, 20551, USA

ARTICLE INFO

JEL classification:
E43
G12
G13
G23

Keywords:
Treasury markets
Hedge funds
Limits to arbitrage
Financial stability
Liquidity

ABSTRACT

This paper studies hedge funds' arbitrage positions in the Treasury cash-futures basis trade, which profits from the disconnect between cash and futures prices. At times, the trade has surpassed $1 trillion in gross exposures. Basis traders consistently account for more than 60% of all hedge fund Treasury positions and 70% of all hedge fund repo. We show how frictions can introduce a positive association between arbitrage quantities and spreads, and how these frictions may propagate stress in the Treasury market during periods of instability such as in March 2020.

1. Introduction

During the middle of March 2020, Treasury markets faced unprecedented disruptions: bid–ask spreads widened to unseen levels; repurchase agreement (repo) rates on Treasury collateral skyrocketed; and various arbitrage spreads diverged. This stress appears to have been due to large sales of Treasuries from multiple entities. In particular, we estimate that hedge funds sold more than $200 billion of cash Treasuries, and reduced total Treasury exposure by $430 billion.[1] These sales followed a dramatic expansion in hedge funds' involvement in the Treasury market: between 2017 and 2019, hedge funds' Treasury exposure increased by almost $1 trillion. After falling substantially in 2020 and 2021, hedge funds again expanded their Treasury market positions by more than $1.1 trillion between June 2022 and September 2023, the last date in our sample.

In this paper, we study the significant time-variation in hedge fund Treasury positions. We use a mix of public and regulatory data to show that a single arbitrage trade — the Treasury cash-futures basis trade — is responsible for most of this variation, and nearly half of hedge fund Treasury sales in March 2020. The "basis trade" profits from the disconnect between Treasury cash and

☆ Views and opinions are those of the authors and do not necessarily represent the views of the Office of Financial Research, the U.S. Department of the Treasury, or the Board of Governors of the Federal Reserve System. The authors thank the editor and associate editor, Yuriy Gorodnichenko and Jonathan Wright, and an anonymous referee for particularly helpful comments. We also thank Jaewon Choi, Wenxin Du, Darrell Duffie, Zhigou He, Arvind Krishnamurthy, Prabhala Nagpurnanand, Zhaogang Song, Annette Vissing-Jorgenson, Jiri Woschitz, Haoxiang Zhu and seminar particpants at the Midwest Finance Assocation Conference, NBER Asset Pricing Seminar, University of Maryland Smith School of Business, Syracuse University Whitman School, Federal Reserve Bank of Atlanta, International Monetary Fund, European Economic Association, OFR-FRBC Financial Stability Conference, Microstructure Exchange, Fixed Income Financial Institutions Conference, ECB Money Markets Conference and FRB Short-term Funding Markets Conference for their comments. Additionally, we thank Ron Alquist, Amanda Buckley, Katherine Gleason, Maryann Haggerty, Mathias Kruttli, Samuel Loch, Matthew McCormick, Sriram Rajan, Stacey Schreft, and Ram Yamarthy for their comments and assistance. Initial versions of this Research circulated as OFR Brief no. 20-01 under the title "Basis trades and Treasury market illiquidity.".

* Corresponding author.

E-mail addresses: daniel.j.barth@frb.gov (D. Barth), jay.kahn@frb.gov (R.J. Kahn).

[1] This makes hedge funds the second largest sellers of Treasuries during March after mutual funds ($266 billion) and ahead of foreign official accounts ($178 billion). Only money market funds were net buyers of Treasuries at $231 billion. For more details on sales across institution types, see Vissing-Jorgensen (2021).

https://doi.org/10.1016/j.jmoneco.2025.103823
Received 17 October 2023; Received in revised form 5 August 2025; Accepted 6 August 2025

Available online 14 August 2025
0304-3932/Published by Elsevier B.V.

futures prices. Traders go long the basis by buying the Treasury security, shorting the corresponding Treasury futures, and borrowing in repo using the Treasury as collateral. We explore the determinants of the equilibrium returns and quantities in the trade, as well as the consequences for financial stability.

First, we develop a simple equilibrium model of the cash-futures basis trade, similar to the models presented in Brunnermeier and Pedersen (2008), Greenwood and Vayanos (2014), and Gromb and Vayanos (2018). In the model, an aggregate quantity of Treasury risk must be shared by dealers and unlevered asset managers. Dealers face balance sheet costs of holding Treasuries, and sell Treasuries to hedge funds who arbitrage between cash and futures markets by passing Treasury risk to asset managers via Treasury futures.

The model demonstrates how the Treasury cash-futures disconnect, which we define as the deviation from the frictionless no-arbitrage benchmark, is determined in equilibrium by a host of factors, including dealers' balance sheet costs, asset managers' demand for Treasury futures, and funding costs for hedge funds, along with margins on Treasury futures and haircuts on repo. The model demonstrates that some factors, such as rising asset manager demand, rising Treasury supply, and repo costs that increase with the volume of repo transactions, lead to a *positive* association between arbitrage quantities and spreads. This is in contrast to a textbook Treasury cash-futures arbitrage condition, as considered in Fleckenstein and Longstaff (2020), which would predict that higher arbitrage spreads imply more costly trading frictions, and therefore lower arbitrage volumes. In our model, the positive association between arbitrage quantities and spreads arises because hedge funds serve as a conduit, passing Treasury risk from dealers to asset managers and requiring higher returns to compensate for the increasing scale of that service. The model also predicts how these factors interact to become destabilizing in periods of stress.

We use data from a variety of sources to empirically test the model's predictions. We begin by documenting that during the two recent periods of rising hedge fund Treasury exposures, the arbitrage spread between Treasury cash and futures prices also rose. This is consistent with the factors that drive higher arbitrage quantities also driving higher arbitrage spreads.

Next, we estimate time-series regressions to evaluate which, if any, of the forces specified in the model are statistically associated with arbitrage quantities and spreads. We find that dealer net Treasury exposure is associated with greater basis trade activity. This echoes Fleckenstein and Longstaff (2020), who argue that the cash-futures basis represents "rental costs" for intermediary balance sheet space.[2] However, our results demonstrate that hedge funds, not dealers, hold Treasuries on behalf of agents like asset managers via the basis trade.[3] We show the demand of these other agents and the frictions hedge funds face also matter for basis trade activity. We find that mutual fund demand for long Treasury futures is positively associated with hedge fund basis trade positions, as are repo rate spreads above the IOER. The latter is consistent with repo costs that increase with the scale of repo activity, and differs from Fleckenstein and Longstaff (2020), which predicts higher exogenous repo costs will result in lower arbitrage activity. Instead, our model of arbitrage quantities identifies how multiple factors, including repo spreads, are positively related to arbitrage activity in equilibrium.

We conclude by examining how hedge fund basis trades can introduce fragility into Treasury markets. Previous papers have discussed links between hedge fund activity and systemic risk, including Brunnermeier and Nagel (2004), Chan et al. (2006), Boyson et al. (2010), Ang et al. (2011) and Aragon and Strahan (2012). Through the basis trade, hedge funds serve as imperfect warehouses of Treasuries, holding them on behalf of asset managers and other investors, and funding them in the repo market. Hedge funds may therefore be particularly susceptible to widening arbitrage spreads or worsening financing conditions.

Indeed, we document that during March 2020, margins increased on Treasury futures and repo rates rose, making the trade more expensive and increasing the risk of maintaining the trade through settlement. In response, we find basis traders sold roughly $100 billion in Treasuries. This builds on previous work demonstrating the importance of hedge fund sales during March 2020, including Schrimpf et al. (2025) and Duffie (2020), as well as to contemporaneous findings in Kruttli et al. (2025), who show that hedge funds' internal risk constraints were important even in the absence of binding external financing constraints. He et al. (2022) also discuss the importance of hedge funds in March 2020, but in their model hedge funds participate in arbitrage between different Treasuries, with somewhat balanced short and long positions across maturities.

We are able to demonstrate the effects of March Treasury market dysfunction on basis traders, making use of novel transaction-level repo data and regulatory data on hedge funds. We show that, at the peak of the stress during the week of March 11th, the repo rate spread for large basis traders increased by at least 30%. This constitutes the first micro evidence of hedge fund financing shocks in March 2020. Examining the cross-section of large basis traders we show that tighter financing constraints and higher repo spreads were associated with more sales of Treasuries. We find only weak evidence related to haircuts, indicating the cost of financing rather than the available quantity of financing may have been more important for motivating basis traders to sell Treasuries. Overall, our work confirms that the basis trade was a significant source of sales by hedge funds in March 2020, and was vulnerable to increases in the frictions highlighted by our model.

This study is related to other work on covered arbitrage trades, including Du et al. (2018) and Avdjiev et al. (2019), who show that regulatory constraints on dealers are partly responsible for deviations from covered interest parity. Our work differs in that we directly connect arbitrage spreads to arbitrage activity. Mitchell et al. (2007) show that convertible-bond arbitrage spreads widened significantly in response to large redemptions from convertible-arbitrage hedge funds and during the LTCM crisis. Similarly, we show that more fragile access to financing for hedge funds may be an underlying systemic risk. Additionally, Boyarchenko et al. (2018)

[2] Similarly, Hazelkorn et al. (2023) show that a basis arises between equity index cash and futures prices due to holding costs for dealers.

[3] In a European context, Peizzon et al. (2025) show that the scarcity that resulted from quantitative easing affected the disconnect between bonds and futures through three costs faced by arbitrageurs: transaction costs, security borrowing costs, and the cost of carry.

find the Supplementary Leverage Ratio may have imposed tighter constraints on dealer balance sheets just as Treasury issuance was increasing, consistent with one source we document for the rise in the cash-futures disconnect. Garleanu and Pedersen (2011) show that the CDS-bond basis is tightly linked to margin requirements on CDS and the general availability of credit, consistent with our finding that when margin constraints rose in March 2020, hedge funds reduced their cash-futures arbitrage positions.

The paper is organized as follows. Section 2 develops a theoretical model that highlights the role of hedge funds as warehouses for Treasuries via the cash-futures basis trade. Section 3 describes the structure of the Treasury futures market and the construction of key variables. Section 4 documents empirical evidence of substantial hedge fund basis trade activity. Section 5 evaluates the drivers of Treasury cash-futures arbitrage growth and contractions, including the substantial unwinding of the basis trade in March 2020. Section 6 concludes.

2. A model of limits to arbitrage and the cash-futures disconnect

In this section, we develop a model that formalizes the effects of various frictions on arbitrage activity between Treasury securities and futures prices, referred to as the “cash-futures” basis trade. Our model examines the determination of both the size of this basis and volume of arbitrage of the basis, and explores consequences for Treasury market functioning and stability. The basis trade links Treasury cash and futures markets to repo markets, and connects hedge funds to dealers, asset managers, and cash providers such as money market funds. Linking the aggregate size of the cash-futures basis trade to the frictions and sources of demand involved in the trade is particularly important for assessing both the recent incredible growth in the trade, as well as how the Treasury market disruptions in March 2020 motivated hedge funds to unwind their basis trade positions and sell Treasuries, further contributing to the stress.

2.1. Assets

There are three periods, $t = 1, 2, 3$. There are two financial assets in positive net supply. The first are Treasury notes, which have fixed quantity S, mature at time 3, and pay \$1 at maturity. The price of a note at time t is $P_{N,t}$. The second are Treasury bills, which have price $P_{C,t}$ and pay \$1 in period $t+1$. Bill prices are determined by an exogenous stochastic short-rate process, however an exact specification of this uncertainty is not necessary for our results below.

In addition, there are two financial assets in zero net supply: repo and futures. Futures contracts for the Treasury note, which are transacted at time $t = 1$, guarantee a price of F_1 for the delivery of a Treasury note at time $t = 2$. Repo markets are open in periods 1 and 2, where agents can borrow at a price $B_{R,t}$ to repay \$1 next period, provided they post Treasury notes as collateral.[4]

2.2. Participants

There are four participants in these markets: *dealers*, *asset managers*, *money market funds*, and *hedge funds*. Dealers represent Treasury market dealers, and are the investors from whom basis traders purchase cash Treasuries. Dealers are mean–variance optimizers and divide their initial wealth between holdings of notes and bills. They are endowed with an existing exogenous stock of Treasury note exposure X_t. We assume this inventory of Treasuries that dealers hold cannot be delivered into Treasury futures, and that dealers do not trade in Treasury futures. This assumption is broadly consistent with aggregate futures data, discussed below, which shows that dealer short futures positions have remained low over the sample period.

Money market funds represent the source of repo loans to basis traders. Money market funds invest only in repo and Treasury bills. We further assume that money market funds have an exogenous preference for Treasury bills due to liquidity demand that depends on the amount of bills they hold,[5] which drives the repo rate above the yield on bills.

Asset managers (i.e., mutual funds), are born at $t = 1$ and live for one period. They have access to notes, futures markets, and an outside investment good. They face two key frictions: they cannot borrow against notes in the repo market (and therefore cannot short notes directly) and they have a duration target for which they face a quadratic penalty. Asset managers therefore invest all their net wealth in the outside asset when yields on that asset are sufficiently high relative to notes, preferring to use scarce balance sheet space for the higher-yielding asset. Meanwhile, they use futures to partially fill the resulting duration gap from their benchmark. This setup matches evidence on the use of Treasury futures by asset managers to match index duration.[6] The relative weight they place on this target acts similar to risk-aversion.

Finally, hedge funds span Treasury notes, repo, and futures markets, and arbitrage differences between them. We assume hedge funds are risk-neutral and only hold notes to be delivered into the futures market. This implies their long and short positions perfectly balance.

[4] Bills are not allowed to be posted as collateral for repo in the model, which matches their low usage as collateral in the data.

[5] The modelling device is similar to Krishnamurthy and Vissing-Jorgensen (2012) and Nagel (2016).

[6] See Barth et al. (2024) on the lack of repo borrowing by asset managers.

2.3. Equilibrium

The model environment generates endogenous cash-futures basis arbitrage spreads and quantities. The fixed stock of Treasuries implies a fixed amount of short-rate risk that must be shared between dealers and asset managers. The stock of Treasuries and dealer Treasury holdings defines a "risk-sharing" line, as shown in the top-left panel of Fig. 1. The line represents that as more Treasuries are held by dealers, prices for cash Treasuries will fall as dealers take on more risk, while prices for futures will rise as asset managers take on less short-rate risk. The slope of this line is determined by the relative risk-aversions of dealers and asset managers. Meanwhile, increases in Treasury supply or in exogenous dealer Treasury exposure cause parallel shifts in the risk-sharing line.

Hedge funds, through their arbitrage activity, act as a temporary warehouse for Treasuries, holding them on their balance sheets for other agents and earning compensation for this service. They facilitate risk-sharing between dealers and asset managers by purchasing notes from dealers and shorting the Treasury futures, using repo to fund their cash purchase. This is consistent with how Treasury cash-futures arbitrage is conducted in practice, and we provide evidence of substantial hedge fund Treasury cash-futures basis trade positions in the next section. As a result of this arbitrage, exposure to Treasuries moves from balance-sheet constrained dealers to asset managers with demand for Treasury exposure.

The arbitrage opportunity for hedge funds is defined by the disconnect between cash and futures prices, $P_{N,1}/P_{C,1} - F_1$, which represents the difference between the price of the Treasury note and the present value of the futures price, discounted at the risk-free rate. In a frictionless market, hedge funds can costlessly absorb Treasury balances and borrow at the Treasury bill yield. As a result, hedge funds act to eliminate arbitrage opportunities and trade until $P_{N,1}/P_{C,1} = F_1$. Graphically, equilibrium in the frictionless market is represented by the dashed line in Fig. 1, along which the cash-futures disconnect is zero. In this frictionless equilibrium, risk is shared optimally between dealers and asset managers since the marginal cost of an additional dollar of Treasury exposure for dealers is equal to the marginal cost of an additional dollar of Treasuries held by asset managers. We can think of this situation as "perfect" warehousing by hedge funds.

We introduce two frictions in the model: repo market illiquidity and margin constraints on both repo and futures. To represent margin constraints, we assume that hedge funds are subject to a margin κ representing the sum of the haircut on repo and the initial margin on futures that scales with short-rate volatility. The maximum that hedge funds can invest in the basis trade occurs at $\mathrm{E}\left[P_{N,2}\right] - F_1 = \frac{\phi_S}{\kappa} W_H^1$, where W_H^1 is hedge funds' initial wealth and ϕ_S is asset managers' risk aversion. Beyond this point, hedge funds arbitrage capacity is exhausted, and any additional Treasuries must be held by dealers. To represent repo illiquidity, we assume that repo lenders (money market funds) charge a spread above the bill yield for lending to hedge funds that increases with the amount borrowed. For simplicity, we assume that $P_{C,1} = (1 + \omega q_{N,1}^H) B_{R,1}$, where $\omega > 0$ is a parameter representing lenders' willingness to provide additional borrowing from hedge funds.

These two frictions lead to the "arbitrage capacity" line in panel (a) of Fig. 1. The increasing slope of the line is determined by repo market illiquidity, while the vertical portion of the line is determined by the arbitrage quantity limit imposed by margins. In equilibrium, these frictions make hedge funds imperfect warehouses for Treasuries, and lead to dealers holding more Treasuries than is optimal. This in turn leads to a positive cash-futures disconnect ($F_1 - P_{N,1}/P_{C,1} > 0$) in equilibrium, which can be seen in the graph by the vertical distance from the frictionless equilibrium line. In other words, the positive cash-futures disconnect in the model is a result of two frictions born by hedge funds: the cost of funding through repo and the cost of margins on futures.

Crucially, dealer costs for holding Treasuries affect the equilibrium disconnect by determining the demand for arbitrage capacity, as does asset manager demand, but these effects are indirect through determining the quantity that is offloaded to hedge funds. The equilibrium disconnect is therefore a result of frictions not only on dealers, but also on the non-bank, non-dealer participants in the various segments of the Treasury market.

2.4. Comparative statics

The model, while stylized, identifies factors that generate variation in arbitrage *quantities* as well as spreads, including how the trade may become destabilizing during periods of stress. In Section 5 we examine these factors empirically. In this section, we provide comparative statics to highlight how these factors affect Treasury cash-futures arbitrage activity and spreads.

2.4.1. The supply of treasuries and dealer treasury exposure

First, in panel (b) of Fig. 1, we show the effect of increases in either X_1, the stock of Treasuries on dealer balance sheets, or in the outstanding supply of notes S, which induce a parallel outward shift in the risk-sharing line. For small shifts, as in the shift from line *Risk sharing* to *Risk sharing′*, dealers are forced to absorb a greater quantity of aggregate risk and the prices of Treasuries fall. Dealers pass some of this increase in aggregate risk to asset managers, who in turn drive down the price of futures. Hedge funds, through their trading of the basis, facilitate the transfer of risk from dealers to asset managers, but require a higher cash-futures disconnect as the quantity of risk transferred (and notes held on hedge fund balance sheets) grows. This results from the arbitrage capacity line being steeper than the $\frac{P_{N,1}}{P_{C,1}} = F_1$ "perfect-warehousing" line. Once Treasury supply becomes sufficiently large, as in the shift to *Risk sharing″*, hedge funds reach their arbitrage capacity limits. At this point, hedge funds cannot transfer any more aggregate risk from dealers to asset managers, and any additional aggregate risk must be absorbed by dealers. This is represented by the vertical portion of the arbitrage capacity line. That increased supply of Treasuries can lead to increases in arbitrage activity suggests large issuance following the Tax Cuts and Jobs Act of 2017 is one potential explanation for the dramatic increase in hedge fund basis trades and the cash-futures disconnect.

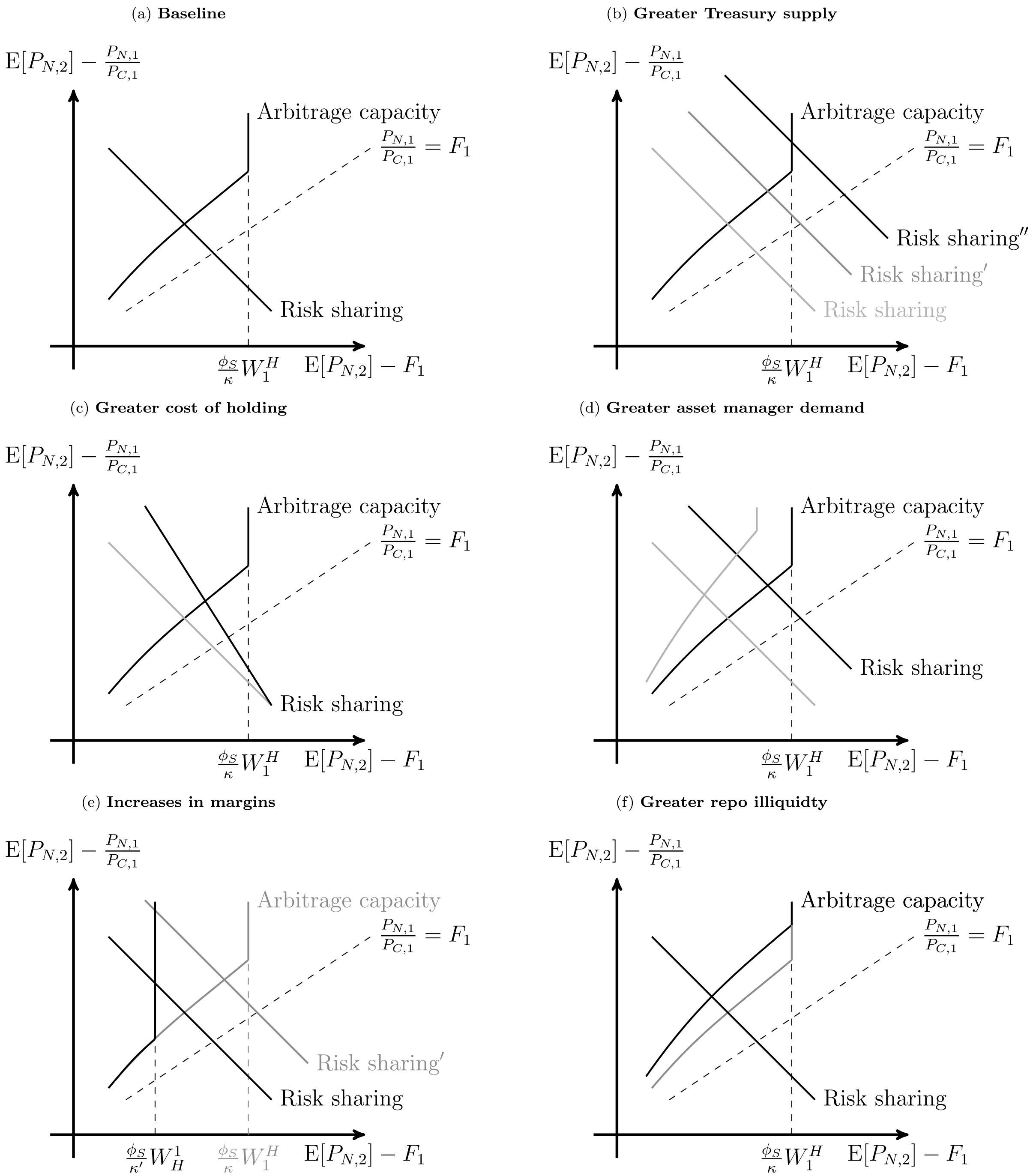

Fig. 1. Comparative statics in the limits to arbitrage model. These graphs display how changing noise trader demand, increases in margins, and repo market illiquidity affect the equilibrium of our limits to arbitrage model.

2.4.2. Dealer balance sheet constraints

Second, in panel (c) of Fig. 1, we examine how increases in dealer costs of holding Treasuries will impact the basis trade. An increased cost of dealers holding Treasuries tilts the risk-sharing curve outward, leading to an increase in the cash-futures disconnect as Treasury risk shifts from dealers to asset managers. This figure points to how increased balance sheet costs from the introduction of the SLR would have affected the basis trade, driving more Treasuries to hedge fund balance sheets and increasing the disconnect.

2.4.3. Asset manager demand

Lastly, in panel (d) of Fig. 1, we examine how increases in asset manager demand for Treasuries impact the basis trade. An increase in asset manager demand (which we interpret as an increase in the duration-gap asset managers need to fill with futures positions) simultaneously shifts the arbitrage-capacity line up and the risk-shifting line in. While the effect on relative prices, $\mathrm{E}[P_{N,2}] - F_1$ and $\mathrm{E}[P_{N,2}] - P_{N,1}/P_{C,1}$, are ambiguous, a rise in asset manager demand for futures leads to an unambiguous increase in asset manager holdings of Treasury futures and in the cash-futures disconnect.[7] The model therefore suggests that increasing demand for long futures positions by asset managers increases both arbitrage activity and the cash-futures disconnect.

2.4.4. Treasury cash-futures arbitrage and financial instability

The model identifies how the basis trade can contribute to financial instability. First, in panel (e) of Fig. 1, we show how higher margins shift the limit on ψ_F inward, as hedge fund arbitrage capacity becomes more limited. For small shifts in margins, there is no effect, as the constraint may not bind before or after. For larger shifts, margins may become binding and basis traders may reduce their positions, shifting a greater proportion of Treasury supply to dealers. As a result, cash prices for Treasuries will fall and the basis will widen. This corresponds to a simple form of the fire-sale effects in standard models of margin constraints such as Brunnermeier and Pedersen (2008). This model is static, but in a dynamic context, the possibility of binding margin constraints in the future could generate a precautionary shift away from the basis trade, so that margin constraints need not be binding today to affect hedge funds' optimal allocation to the trade.

Additionally, the framework shows how repo market frictions can affect the basis trade in panel (f) of Fig. 1. As ω increases, it shifts the arbitrage capacity line out from its initial position. The increase in the cost of repo leads to shifts from asset managers bearing risk to dealers bearing risk as arbitrage becomes more costly. The cash-futures disconnect also rises to compensate hedge funds for this increased illiquidity. Again, while in this static model what matters is repo illiquidity today, in a dynamic model hedge funds would also consider the possibility of illiquidity in the future.

In all cases, limits to arbitrage faced by hedge funds either amplify the effects of greater Treasury exposures or have a direct effect on prices of cash Treasuries. During March 2020, as we will show, both of these risks materialized: real money investors sold Treasuries to dealers, increasing their exposure; margins on futures contracts increased; and repo market illiquidity drove repo rates up.

3. Market structure and construction of key variables

The basis trade links the spot price and futures price of Treasuries. In this section, we first describe the structure of the Treasury futures market, with a focus on market details that are important for correctly calculating prices and arbitrage spreads. We then document the construction of key variables used in the analysis and the dynamics of the disconnect between cash and futures prices relative to the frictionless benchmark. Data sources are provided in Appendix A.

3.1. Structure of the futures market

3.1.1. Trading and delivery

Here, we briefly review some of the details of the Treasury futures market, though we also refer the reader to more in-depth treatments such as Burghardt and Belton (2005). The Treasury futures market is operated in the United States by the Chicago Board of Trade (CBOT). In a futures contract, the short position agrees to sell (deliver) the security to the long position at a future date, at a price agreed upon today. Treasury futures contracts, unlike other interest rate futures, require physical delivery of an underlying Treasury. The *option* to carry these contracts into delivery is ultimately what enforces the arbitrage relationship between cash and futures prices of Treasuries. However, actual delivery is rare, and positions are usually rolled over prior to the beginning of the delivery month.[8]

The CBOT offers Treasury futures contracts at various maturity points, commonly referred to as 2-year notes, 5-year notes, 10-year notes, 10-year ultra notes, bonds, and ultra bonds. To keep the contracts suitably liquid, any Treasury among a predefined "deliverable set", based both on the Treasury's original maturity and its residual maturity on the last day of the delivery month, may be delivered to settle the contract.[9] In an effort to make all Treasuries in the deliverable set for a particular contract comparable, the CBOT establishes "conversion factors" that are applied to the Treasury futures price, and are determined by the coupons and remaining maturity of the Treasury security. since these factors are not based on market prices, at any given time one deliverable Treasury, referred to as the "cheapest-to-deliver" (CTD), will be the most desirable for the short position to deliver.[10] Market efficiency virtually guarantees the convergence of cash and futures prices by the delivery date, because on that date, the Treasury can be delivered directly into the short contract, receiving the futures price. As a result, cash and futures prices converge by the delivery date, but *only* for the CTD.[11]

7 This ambiguity is shown formally in Appendix B.3.3.

8 See Appendix C. Within the delivery month, several options are available for short positions regarding the timing of delivery and the exact Treasury delivered. These options are reviewed in Burghardt and Belton (2005).

9 Table C.1 in the Appendix provides details on the deliverable sets and terms for each of these Treasury futures.

10 We describe our methodology for identifying the CTD security in Appendix C.2.1.

11 This is shown in Appendix C.3.

3.2. Construction of the empirical cash-futures disconnect

We now establish an empirical analogue to the cash-futures disconnect in our model. This requires a multi-period analogue to the arbitrage spread in the two-period model, which we now derive. An investor can either purchase the Treasury today in the cash market at a price $P_{t,\tau}$, or can purchase a Treasury futures contract today that will deliver the same Treasury at time T for a price $F_{t,\tau,T}$.[12] In a frictionless market, the only difference between the cash price and the futures price of a Treasury should be that the investor has to wait for the futures contract to deliver and forgoes the value of any intervening coupons (c_s). Discounting these cash flows to the present using the price of a risk-free bond, $B_{t,s}$, we have:

$$P_{t,\tau} = \sum_{s=t}^{T} B_{t,s} c_s + B_{t,T} F_{t,\tau,T}. \tag{1}$$

Equivalently, Eq. (1) can be restated in terms of yields:

$$y^F_{t,\tau,T} = \left(\frac{F_{t,\tau,T}}{P_{t,\tau} - \sum_{s=t}^{T} B_{t,s} c_s} \right)^{\frac{252}{T-t}}. \tag{2}$$

The yield $y^F_{t,\tau,T}$ is known as the *implied repo rate* (IRR). The yield-equivalent no-arbitrage condition is $y^F_{t,\tau,T} = r_{t,T}$, which states that the yield-to-delivery on a portfolio that is long the cash Treasury note and short the futures must be equal to the yield on an equivalent-maturity bill.[13]

Consistent with the model developed in Section 2, we refer to the difference between $y^F_{t,\tau,T}$ and $r_{t,T}$ as the *cash-futures disconnect.*[14] The magnitude of the disconnect therefore reflects both arbitrage opportunities and the severity of limits to arbitrage, as outlined in the model and discussed below.

Fig. 2 plots the cash futures disconnect, defined as the implied repo rate minus an equivalent-maturity Treasury bill yield, over time. For each futures contract (i.e. the 2-year contract, 5-year contract, etc.), we use the contract with the highest volume for each day to avoid noise introduced by traders rolling into the next-to-deliver contract. We follow this convention through the remainder of the paper. Fig. 2 shows that the difference between the implied repo rate and equivalent-maturity Treasury bill yield generally hover around zero, suggesting that much of the difference between cash and futures prices is explained by risk-free rates; when risk-free rates are higher, the prices of cash bonds (which are in present dollars) are discounted more heavily compared to futures prices (which are in dollars at the delivery date). However, the figure also shows that at times, material and persistence deviations from the frictionless no-arbitrage condition do exist.

4. Hedge funds and the treasury cash-futures basis trade

Section 2 developed a model in which arbitrageurs serve as a conduit, passing aggregate Treasury risk from dealers to asset managers via cash-futures arbitrage. Treasury cash-futures arbitrage in practice follows closely the mechanics of the arbitrage trade outlined in Section 2. When the cash Treasury price is too low relative to the futures price, arbitrageurs go “long the basis” by buying the cash Treasury note, shorting the Treasury futures contract, and borrowing in the repo market to finance the purchase of the Treasury note. This trade is commonly known as the “Treasury cash-futures basis trade”, or simply the “basis trade”.

In this section, we provide evidence that during two recent periods, hedge funds built up substantial Treasury cash-futures basis trade positions. The scale of the positions formed in 2018–2020 likely served as a vulnerability that magnified stress during March 2020, as specified by the model, and is the focus of Section 5.3. Here, we detail the significant arbitrage quantities implied by hedge fund portfolios.

4.1. Aggregate evidence of hedge fund basis trade activity

The top-left panel of Fig. 3 shows that beginning in early 2018, and again in mid-2022, traditional asset managers accumulated substantial long Treasury futures positions. Between December 2017 and December 2019, the notional value of long Treasury futures positions held by asset managers grew from \$487 billion to \$888 billion, an increase of 82%. During the stress in early 2020, these positions fell significantly, reaching a low of \$611 billion in November of 2021. Beginning in June of 2022, asset manager long Treasury futures again began to rise rapidly. In June 2022, asset manager longs were \$709 billion, and reached \$1.14 trillion by the end of September 2023, a growth of 61%.

[12] Specifically, $F_{t,\tau,T}$ is the invoice price, or “cash-equivalent” price, for the Treasury futures contract agreed to at t and delivering at time T. The invoice price takes account of conversion factors that are intended to make bonds in the deliverable set roughly equivalent.

[13] Consistent with our model's frictionless baseline, this no-arbitrage condition assumes not only that hedge funds earn zero profit, but also that money market funds are indifferent between term repo lending and holding a bill, and that dealers pass-through their lending to hedge funds without an additional spread. In practice, hedge funds typically face a spread above the bill rate in repo markets, however it is difficult to directly measure hedge fund term repo rates, as available data are either limited to overnight transactions or reflect pricing in interdealer markets rather than transactions involving hedge funds. We discuss repo spreads in more detail below.

[14] The cash-futures disconnect differs from the cash-futures basis in that the basis is defined as the difference between the implied repo rate and the available market repo rate.

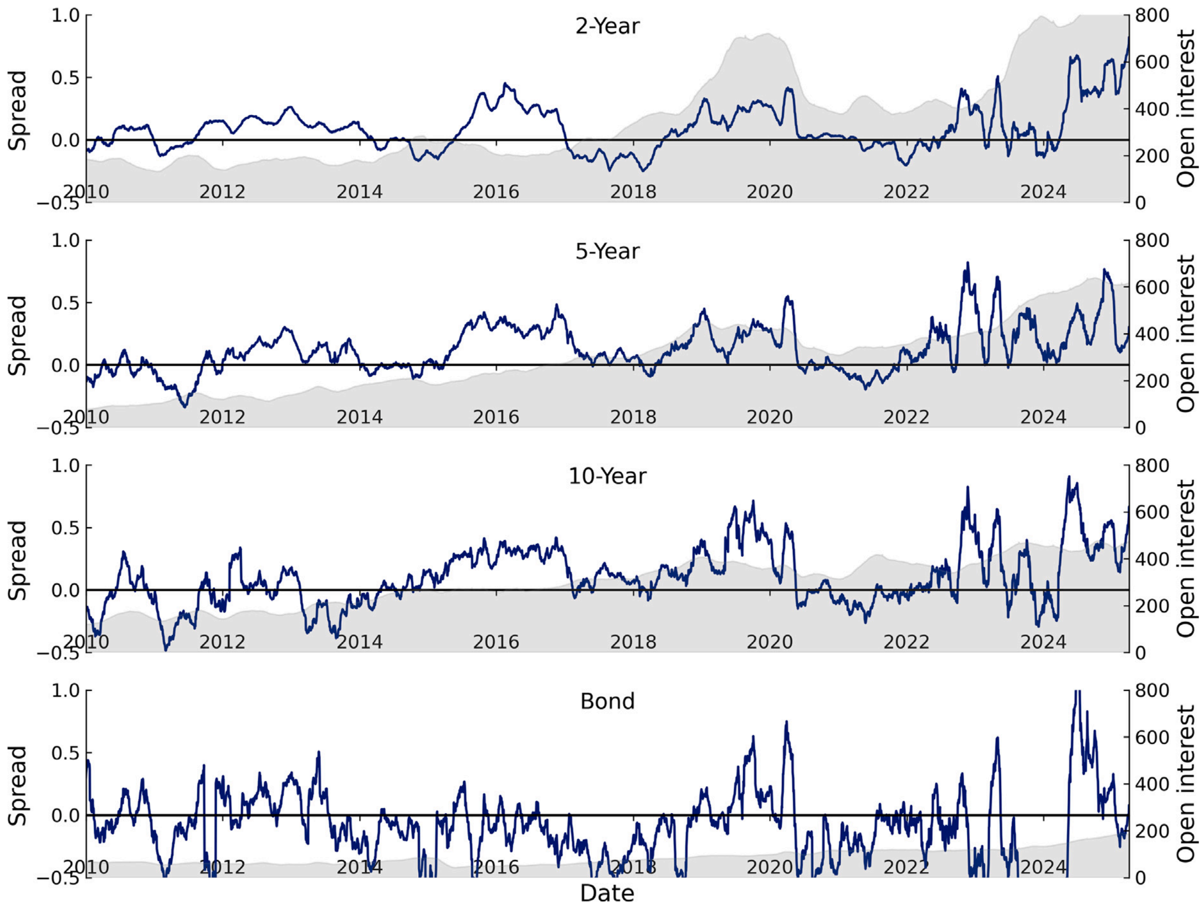

Fig. 2. Deviations of Treasury prices from the replicating portfolio. Each series graph shows the deviation of the implied repo rate from the yield on an identical maturity bill and the open interest in that contract. Spreads are in percentage points, while open interest is in billions of dollars of notional value. The implied repo rate is the return on a given day to buying the cash Treasury and holding it to delivery into a short futures contract entered into on the same day. Where a bill of identical maturity to the delivery date of the contract cannot be found, the price of a bill is interpolated from surrounding bill prices. Values above zero imply a replicating portfolio of bills and futures is overvalued relative to the cheapest-to-deliver cash security. These series use the second-to-deliver contract, the contract with a delivery date following the nearest available delivery date.

Also shown in the top-left panel of Fig. 3 is that during both periods, hedge funds met almost the entirety of this growth. Between December 2017 and December 2019, hedge fund short futures positions grew from $335 billion to $746 billion, an increase of $411 billion, more than covering the $401 billion increase in long futures held by traditional asset managers. By June of 2022, hedge fund short futures had fallen to $286 billion, then grew to $1.04 trillion by September 2023, an increase of 263%. Of this growth, the vast majority was concentrated in the 2-year, 5-year, and 10-year contracts, those most likely to be associated with cash-futures basis trades.[15] By contrast, dealers met very little of asset managers' demand for long Treasury futures. Dealers increased their short futures positions by only $64 billion between December 2017 and December 2019, and by only $71 billion between June 2022 and September 2023.

Next, we turn to hedge fund balance sheet information from Form PF. Coincident with the increase in hedge fund short futures positions, from 2018 through 2019 total hedge fund Treasury exposure, calculated as the sum of (absolute) long and short notional positions (including both cash holdings and derivatives) also increased significantly. The top-right of Fig. 3 shows that in December 2014, total hedge fund Treasury exposure was $918 billion. This is reflected by the sum of total short and total long Treasury positions. By the end of 2017, this exposure was $1.16 trillion, and by December 2019 had grown to $2.19 trillion. Consistent with the futures data, hedge funds' aggregate Treasury exposure fell beginning in the first quarter of 2020, reaching a post-2020 low of $1.52 trillion in June of 2022. By September 2023, Treasury exposure rose to an all-time high of $2.64 trillion. These patterns are consistent with a significant growth in the basis trade. The growth in short Treasury exposure reported in Form PF aligns closely

[15] This due to the relatively smaller margins on shorter-maturity contracts and their larger open interests. The growth in 2, 5, and 10 year hedge fund short Treasury futures was $352 billion between December 2017 and December 2019 (86% of the growth over this period), and $694 billion between June 2022 and September 2023 (92% of the growth over this period).

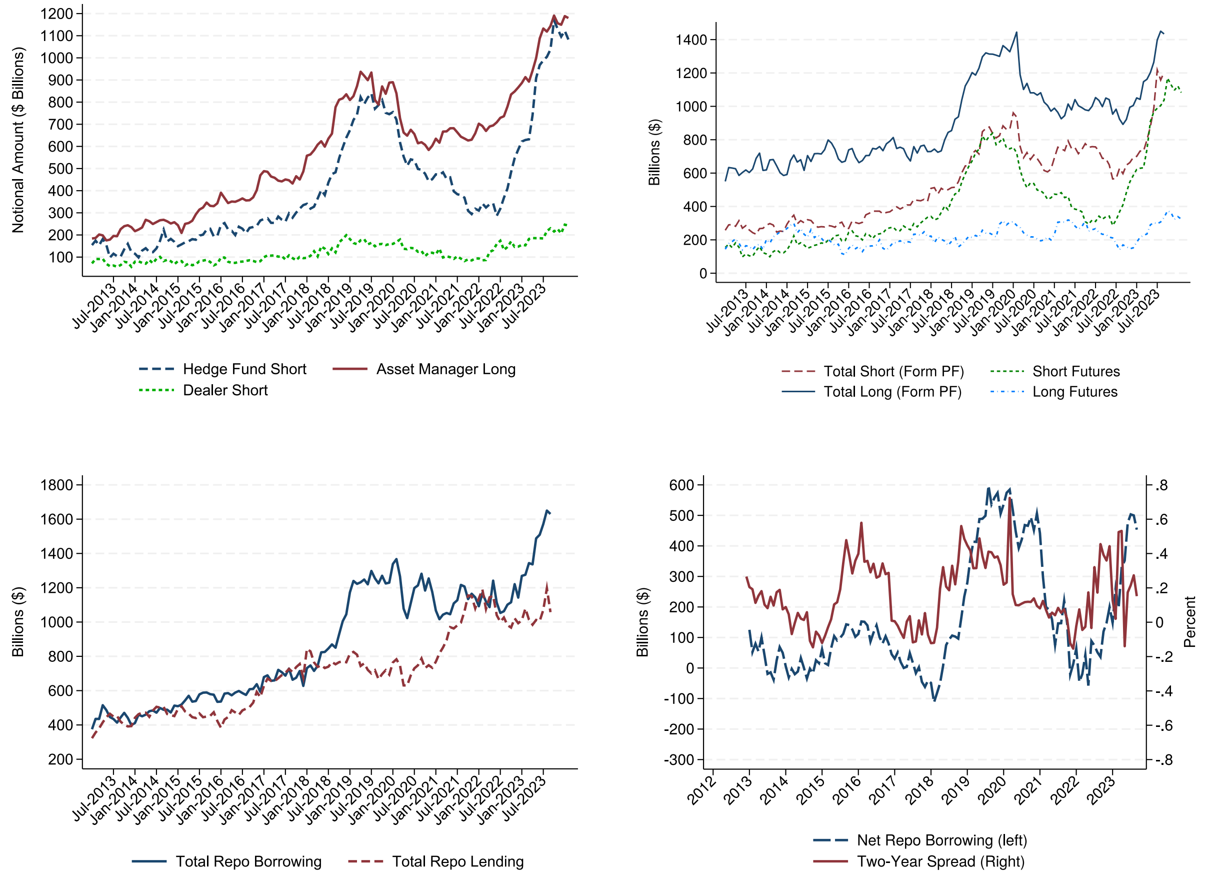

Fig. 3. Futures contracts of hedge funds, dealers, and asset managers. The top-left panel of the figure shows notional billions of dollars in long Treasury futures positions for asset managers and short Treasury futures positions for dealers and hedge funds, aggregated across all contracts. The top-right panel shows aggregate hedge fund long and short Treasury exposures alongside their notional long and short Treasury futures exposures in billions of dollars. The bottom-left panel shows aggregate hedge fund repo borrowing and lending. The bottom-right panel shows hedge fund aggregate net repo positions, defined as aggregate repo borrowing minus lending, against the cash-futures disconnect of the 2-year futures contract.

with the increase in short futures positions from CFTC data. However, the increase in long Treasury exposures on Form PF appears to come almost exclusively from cash securities; while long Treasury exposures were growing quickly on Form PF, long futures positions from CFTC data remained relatively flat.

Further evidence of large basis trade positions is found in hedge funds' repo activity. Because the basis trade requires only the long note position be financed in the repo market, with no corresponding repo lending associated with the short leg of the trade, an increase in the basis trade would increase repo borrowing but not repo lending.[16] The bottom-left panel of Fig. 3 shows that, indeed, during both periods of rising short futures positions, hedge fund net repo borrowing, calculated as the difference between repo borrowing and lending, was also growing. By December 2019, aggregate net repo had grown to $536 billion, actually turned negative at various points in 2021 and 2022, then rose again to $573 billion by September 2023. The bottom-right panel of Fig. 3 shows hedge funds' aggregate net repo borrowing closely tracks the two-year cash-futures arbitrage spread.[17]

4.2. *Large basis traders*

4.2.1. *Large basis trader classification*

To study cross-sectional variation in basis trade activity, we focus on a set of funds with large basis trade positions. For each month in the two basis trade periods of January 2018–December 2019 and June 2022–September 2023, we define a hedge fund

[16] This differs from other types of cash Treasury arbitrage trades, such as the on-the-run/off-the-run trade, which would include both a repo and a reverse repo.

[17] While hedge fund short futures remained relatively flat during 2015–2017, despite rising net repo positions, in unreported results we find that the net percentage of short Treasury exposure comprising short futures rose during this period, suggesting a reallocation away from other Treasury arbitrage trades and towards the cash-futures basis trade.

Table 1
Hedge fund treasury and repo exposure. This table shows hedge fund long, short and net Treasury exposure and repo borrowing and lending exposures from Form-PF for several key dates, as well as changes in exposure from the end of February to the end of March, 2020 and estimated cash sales over that month. Additionally, the table includes Treasury futures for all funds derived from the CFTC's Traders in Financial Futures data (using the last reported date in the month), and estimated Treasury cash positions (total Treasury exposure less Treasury futures). For total Treasury exposure and estimated cash exposure, sales are estimated by using the ICE Treasury index for valuation adjustment.

	Levels						Changes
	2017 Dec	2019 Dec	2020 Feb	2020 Mar	2022 Jun	2023 Sep	2020 Feb–March
All hedge funds							
Treasury exposure							
Long	728.6	1327.1	1445.0	1189.7	953.1	1434.0	−255.3
Short	430.3	863.3	936.0	760.7	565.2	1205.6	−175.2
Treasury futures							
Long	219.1	306.3	286.6	275.3	211.5	334.0	−11.3
Short	335.2	745.5	720.8	618.3	286.4	1037.7	−102.5
Treasury cash sales							
Long							−294.59
Short							82.16
Repo							
Borrowing	627.5	1229.6	1367.4	1265.7	1113.5	1630.4	−101.7
Lending	676.3	693.5	782.9	746.8	1040.0	1057.8	−36.1
Large basis traders							
Treasury exposure							
Long	339.2	832.0	892.3	734.3	646.0	995.4	−158.0
Short	236.2	600.6	663.0	571.3	426.0	833.5	−91.7
Repo							
Borrowing	493.3	1064.8	1153.6	1057.2	983.4	1509.9	−96.4
Lending	534.7	577.9	633.3	632.8	951.3	968.4	−0.5

as a "large basis trader" if the fund is in the top 50 of all funds in both aggregate Treasury exposure and net repo borrowing in that month and year. This produces 41 unique funds that were classified as large basis traders at some point during the 2018–2019 period, with an aggregate net asset value of $168 billion as of December 2019, and 54 unique funds in the 2022–2023 period, with aggregate net asset value of $337 billion as of September 2023. There are 25 funds that are large basis traders in both periods, which tend to comprise the largest basis traders; 17 out of the top 20 large basis traders in 2018–2019, ranked by total Treasury exposure in December 2019, were also large basis traders at some point in 2022–2023.[18]

Large basis traders are responsible for much of the aggregate Treasury exposure and repo activity of hedge funds in the Form PF data. Figure D.8 plots aggregate long and short Treasury exposure as well as net repo borrowing for all hedge funds and for large basis traders only. These numbers are also reported in Table 1. In December 2019, the fraction of total hedge fund Treasury exposure attributable to large basis traders was 68%, and was 64% in September 2023. The fraction of total repo positions attributable to large basis traders were 85% and 92% on these dates, respectively. These figures imply a high degree of concentration in overall Treasury and repo exposures, due largely to large basis traders, given that large basis traders only accounted for 8.2% of total non-zero repo borrowing observations in December 2019 and 6.3% of total non-zero Treasury exposure observations. The magnitude of repo borrowing by basis traders in 2019 would be equivalent to more than 56% of all repo lending by primary dealers, and basis trader net repo borrowing would have been equivalent to around 25%. As shown in Table 1, net repo borrowing for large basis traders grew by $529 billion between December 2017 and December 2019 and by $510 billion between June 2022 and September 2023. These provide additional estimates of the overall size of hedge fund basis trade activity.

5. Empirical evidence on basis trade expansions and contractions

In this section, we empirically test the model's predictions that specific factors affect the scale of hedge fund basis trade activity and the size of the disconnect between Treasury cash and futures prices. We provide both time-series results for the period 2010–2023 and a case study of the substantial unwind of hedge fund basis trade positions in March of 2020.

The model highlights five specific factors that could affect the scale of hedge fund basis trade activity: the total supply of Treasuries, asset managers' demand for long Treasury futures, the balance sheet Treasury holdings of dealers, repo financing costs, and margins on Treasury futures. In this section, we focus on plausible proxies for ceteris paribus effects, which are the empirical counterparts to the comparative statics provided in Section 2.4. In the Appendix, we also provide evidence on general equilibrium associations.

[18] In Appendix D.1 we consider alternative definitions of large basis traders, and find that our results are robust to these alternative definitions. We also discuss concentration among large basis traders in Appendix D.2.

Table 2
This table reports regression results of arbitrage quantities and spreads on empirical proxies for the factors identified in the model. Variables are defined in Section 5. Weekly regressions use Newey–West standard errors with 30 lags, while quarterly regressions use Newey–West standard errors with 2 lags. Statistical significance: ***p < 0.01, **p < 0.05, *p < 0.1.

PANEL A: Leveraged funds short treasury futures (Weekly)					
	(1)	(2)	(3)	(4)	(5)
FNMA MBS OAS	0.028**				0.015**
	(0.014)				(0.007)
Dealer net treasury exposure		2.777***			1.984***
		(0.364)			(0.471)
MMF repo capacity			0.264***		0.131***
			(0.046)		(0.040)
Futures margin				−1.045	−0.948
				(1.911)	(1.537)
Quarter end					0.199
					(0.268)
R-squared Adj.	0.076	0.577	0.000	0.464	0.700
No. observations	745	783	783	744	706
PANEL B: Leveraged funds short treasury futures (Quarterly)					
	(1)	(2)	(3)	(4)	(5)
Mutual fund treasury gap	4.351***				1.776*
	(1.087)				(0.986)
Dealer net treasury exposure		2.874***			1.398**
		(0.356)			(0.588)
MMF repo capacity			0.222***		0.118***
			(0.049)		(0.043)
Futures margin				−4.630**	−0.392
				(2.314)	(1.846)
R-squared Adj.	0.531	0.571	0.382	0.035	0.691
No. observations	48	48	48	48	48
PANEL C: Cash-futures arbitrage spread (Weekly)					
	(1)	(2)	(3)	(4)	(5)
FNMA MBS OAS	0.000				0.001
	(0.001)				(0.001)
Dealer net treasury exposure		0.094***			0.196***
		(0.029)			(0.038)
MMF repo capacity			−0.008		−0.020***
			(0.005)		(0.004)
Futures margin				0.135	0.469*
				(0.258)	(0.247)
Quarter end					0.004
					(0.120)
R-squared Adj.	−0.000	0.043	0.001	0.029	0.167
No. observations	744	753	753	714	705

5.1. Time-series evidence: Arbitrage quantities

We begin by examining hedge fund short Treasury futures scaled by outstanding notes and bonds (in percentage terms) as a measure of arbitrage quantities. We test model predictions using both weekly and quarterly data, depending on availability. Weekly data offer more power, while quarterly data allow more direct measurement of asset manager demand.

At the quarterly frequency, in order to proxy for asset manager demand, we are motivated by Barth et al. (2024) in constructing a "mutual fund Treasury gap": the difference between core bond mutual fund Treasury weights and those in the Bloomberg Aggregate index, scaled by core bond fund assets under management divided by outstanding notes and bonds. As in the model, a wider gap, often reflecting shifts into MBS amid higher returns relative to Treasuries, implies greater demand for long futures to meet duration targets. Since this data is only available quarterly, at the weekly level we instead proxy for demand using the MBS option-adjusted spread (OAS).

For our other frictions, the same variables are used at a weekly and monthly frequency. Dealer Treasury holdings scaled by outstanding notes and bonds proxy for X_t, the net exposure held by dealers. To capture variation in repo funding conditions, we construct a measure of "MMF repo capacity": government money market fund (MMF) assets net of their Treasury bill holdings, scaled by outstanding notes and bonds. This measure reflects the supply of cash available to repo and is largely driven by factors like MMF inflows and bill issuance–variables that are unlikely to be directly influenced by basis trade activity.[19] As such, it serves as a proxy for the availability of cheap repo financing that is plausibly independent of basis trader demand. For futures margins,

[19] See Hempel et al. (2023) on the relationship between MMF flows, bill issuance and total repo.

we use the end-of-period level relative to its rolling 90-day average, filtering out movements that are linked to Treasury volatility or slow-moving changes in exposures, rather than idiosyncratic decisions by the CBOT.[20]

Results for regressions using weekly data are reported in Panel A of Table 2. In columns (1) and (2) we examine factors related to the transfer of Treasury risk from dealers to asset managers. Column (1) shows that, as predicted by the model, during periods of high mutual fund demand for Treasury futures as proxied by the MBS OAS, Treasury cash-futures arbitrage is higher. The coefficient is significant at the 5% level, and implies that a 100 basis point change in the OAS associates with an 2.80 percentage point change in short hedge fund Treasury futures (scaled by outstanding notes and bonds). In column (2), we examine dealer Treasury holdings. As Section 2.4 shows, changes to the aggregate supply of Treasuries (S_t) or the exogenous stock of Treasuries dealers hold (X_t) both shift the risk-sharing line outward, and therefore have similar consequences for arbitrage quantities (and spreads). In column (2), we focus on dealer balance sheet holdings of Treasuries scaled by outstanding notes and bonds.[21] Similarly, column (2) shows that higher net dealer Treasury exposure is also associated with higher arbitrage quantities. The coefficient is significant at the 1% level, and implies a 1% higher Treasury exposure of dealers associates with 2.77% higher short hedge fund futures.

In columns (3) and (4) we examine factors related to the costs of trading the basis. In column (3), we find that MMF assets — the proxy for available cash in the repo market — is strongly positively associated with arbitrage quantities, consistent with a greater availability of repo cash making repo funding cheaper. A one percentage point increase in MMF assets increases hedge fund short futures by 0.26 percentage points.[22] Similarly, in column (4), we find that higher futures margins are negatively related to arbitrage activity, though the coefficient is statistically insignificant.

In column (5), we include all these factors together in a single specification. While coefficient magnitudes generally shrink, the coefficients on MBS OAS, net dealer Treasury holdings, and money market fund repo capacity all remain statistically significant and economically important, and the coefficient on margins remains negative, further supporting that each of these measures represent independent sources of variation affecting hedge fund cash-futures basis positions.

Panel B of Table 2 estimates a similar set of specifications to Panel A but at a quarterly frequency. The main difference is the measure of asset manager futures demand, which can be more directly estimated by the "mutual fund Treasury gap", as defined above. The other variables are measured analogously to Panel A. In column (1), we find that the mutual fund Treasury gap is strongly related to hedge fund short futures positions, with a t-statistic near four. A one percentage point larger gap between mutual fund Treasury holdings and the percentage holdings in the benchmark associate with 4.35 percentage points larger hedge fund short futures positions. Columns (2) and (3) provide similar findings to those in Panel A. However, in column (4), futures margins are now statistically significant at the 5% level, providing stronger evidence that higher margins reduce basis trade activity. As with Panel A, including all these factors together somewhat reduces magnitudes and, in the case of margins, the statistical significance. Nonetheless, together Panels A and B provide strong support for the model's predictions.

5.2. Time-series evidence: Arbitrage spreads

In Panel C of Table 2, we return to the weekly frequency but now focus on the cash-futures arbitrage spread as the left-hand side variable. We use the spread for the two-year contract because that contract has the largest amount of hedge fund short positions, though results are similar for the five-year spread. The variables are the same as in Panel A. In columns (1)–(4), we find the signs on the coefficients are consistent with the model, but only dealer net Treasury exposure is statistically significant. However, once all factors are included together in column (5), the coefficients on dealer Treasury holdings, money market fund assets, and futures margins are all significant at either the 1% or 10% levels and have the predicted signs. Only the MBS OAS coefficient remains insignificant, likely due to the relatively large noise in the OAS.[23]

In total, across both weekly and quarterly regressions, we find substantial empirical evidence that the factors identified in the model are important for determining both arbitrage quantities and spreads. Importantly, the adjusted R^2 reported in column(5) in all three panels of Table 2 indicates that the empirical proxies of the factors identified in the model explain a significant amount of the time-series variation in arbitrage quantities and spreads. Next, we turn to an assessment of these factors in March 2020, when the Covid-19 shock quickly and significantly disrupted Treasury markets, raising both repo costs and margins, and flooding dealers' balance sheets with Treasuries, and show how these rapid changes motivated a substantial unwind of hedge fund basis trade positions.

[20] See Figure F.14 in the Appendix, which shows how margins are influenced by Treasury volatility.

[21] We focus on dealer Treasury holdings because once scaled by aggregate Treasuries outstanding, the series is stationary, which is necessary because the other variables in Table 2, including the left-hand side measure of arbitrage quantities, are also stationary. Conversely, aggregate Treasuries outstanding are not stationary, so cannot be used as an independent variable in Table 2. In Figure E.13 of Appendix E, we plot the aggregate supply of Treasuries against aggregate hedge fund short futures positions. The series comove tightly, and since 2015 are 85% correlated.

[22] In Table E.2 of the Appendix, we show that using the actual repo rate (minus the risk-free rate) is *positively* related to arbitrage quantities. This is also consistent with the equilibrium predictions of the model, which indicate that higher hedge fund repo quantities lead to higher repo rates. Also in that Table, we show that quarter end effects are unrelated to basis trade quantities and spreads, unlike in Du et al. (2018) or He et al. (2022), who find that quarter-end effects are important for CIP deviations.

[23] Indeed, as shown in column (1) of Table E.2 in the Appendix, when the mutual fund Treasury gap is used as the measure of mutual fund demand in quarterly regressions, the coefficient is positive and statistically significant at the 10% level.

5.3. The basis trade unwind in march 2020

A key prediction of the model is that during times of stress, heavy activity in the basis trade can serve as an amplifier of the stress caused by sales of Treasuries by other investors, which increases both the amount of Treasuries held by dealers and the volatility of Treasury prices. In March 2020, selling pressure by real money investors, particularly foreign central banks and domestic mutual funds, placed pressure on dealers' balance sheets.[24] As shown in the top panel of Fig. 5, these sales by foreign official accounts, which were likely undertaken to build up dollar buffers for foreign central banks, coincided with the initial stress in Treasury markets. Without immediate buyers, these Treasuries remained on dealers' balance sheets and made dealers hesitant to create markets in off-the-run Treasuries.

The crisis directly increased the risks that hedge funds face through the basis trade, raising margins and increasing spreads in the repo market. As a consequence of dealers' inability to make markets, volatility spiked and bid–ask spreads widened.[25]. Further, rising volatility led to increased margins on Treasury futures. The top panel of Fig. 5 shows that margins began to increase just before hedge funds began to shrink their short futures positions. Further, the variation margin payments on the futures contract were not fully offset by the increase in prices in the long note position.[26]

Beginning in late February, stress in the repo market began to materialize as well. Fig. 4 illustrates the pass-through of federal funds rate target changes onto repo rates for sections of the "Delivery versus payment" (DVP) market. We focus on this market because it is the only available source of transaction-level data on hedge fund repo. In the top panel, we present the spread of DVP rates over the federal fund's target rate. Prior to March, these facilities provided tight control over rates in the inter-dealer and sponsored lending sections of the market. The repo facility gives dealers an outside option from which to borrow, and when effective, dealers in the interdealer market will not accept higher rates than in the RP facility. Similarly, the RRP facility gives an outside option to money-market funds and other sponsored lenders, who can lend to the Federal Reserve overnight and receive the RRP rate. In early March 2020, liquidity became scarce and volume in the RP facility increased, as shown in the bottom panel of Fig. 4. As volumes in the facility increased, dealers became concerned about the aggregate cap on the facility, and the weighted average rate on repo was driven up beyond the minimum in auctions. The grey shaded regions identify periods when the rate was above this minimum bid. During these periods, interdealer rates in DVP also rose beyond the upper bound, as dealers who were concerned about funding from the RP facility being unavailable in the afternoon were willing to pay higher rates to secure funding in the morning.

5.4. Repo market stress and large basis traders

Stress in the repo market disproportionately affected hedge funds. Using data on all repo transactions in the DVP market for the period October 2019 to May 2020, we empirically assess the repo rate spreads hedge funds paid during the March 2020 instability. We estimate regressions of the form:

$$\text{repo rate}_{i,j,s,t} = \alpha_{i,t} + \rho \times \text{collateral}_{s,t} + \beta \mathbf{1}_{j\in \text{LBT}} + \delta \mathbf{1}_{j\in \text{LBT}} \times \mathbf{1}_{t\in \text{repo stress period}} + \epsilon_{i,j,s,t}, \tag{3}$$

where i denotes the lender identity, j denotes the borrower identity, s denotes the underlying security used as collateral, and t denotes the hour and date in which the transaction occurs.[27] On the right hand side, $\alpha_{i,t}$ is either an hourly fixed effect or a lender-by-hour fixed effect, collateral controls are security and time specific variables designed to control for special collateral rates, $\mathbf{1}_{j\in \text{LBT}}$ is a dummy variable equal to one if the borrower was classified as a large basis trader in the final quarter of 2019, and $\mathbf{1}_{t\in \text{repo stress period}}$ is a dummy equal to one if the repo facility on that given day was bid up beyond the minimum rate.[28] This specification allows us to examine the rates that large basis traders pay relative to the rates that other borrowers from the same lender received in the same hour while controlling for the effects of specific collateral.

Results are shown in the top panel of Table 3. The first four columns include day-by-hour fixed effects, separately interacted with indicators for whether the security is on-the-run, is the cheapest-to-deliver for any Treasury futures contract, or the security is borrowed on the same day from the System Open Market Account (SOMA) facility.[29] During non-stress periods, large basis traders pay 9–10 basis points higher rates than the rates on other transactions. These spreads widened by 2.6–4.4 basis points during stress periods, depending on the specification.

The final two columns of Table 3 take an extreme view of these collateral-specific rates by using spreads over rates in the same hour for the same collateral in the blind-brokered segment of DVP, which are between clearing members of FICC such as large dealers and banks. This necessarily restricts the sample to collateral which trades for both large basis traders and dealers in the same hour. Even so, we still find that large basis traders pay more than 7 basis points higher rates, with the spread increasing by

[24] Duffie (2020) and Vissing-Jorgensen (2021) provide broad overviews. Sales by domestic mutual funds have been highlighted by Pástor and Vorsatz (2020) and Vissing-Jorgensen (2021). Sales by foreign official investors are discussed in Weiss (2022) and Alquist et al. (2022).

[25] We review this evidence in Appendix G

[26] Avalos and Sushko (2023) document a similar pattern between Treasury futures margins and hedge fund short futures positions in March 2020.

[27] We form a panel of all repo transactions in the DVP market aggregated to the level of the lender by counterparty by collateral by transaction time in ten minute buckets.

[28] These periods include all of March 3–5 and March 9–12 2019 as well as prior to 1:00 PM on March 16, since the first repo offering on March 16 at 8:45 was oversubscribed and the second offering a 1:45 was undersubscribed.

[29] These aspects are shown to be important determinants of DVP repo rates in Hempel and Kahn (2021).

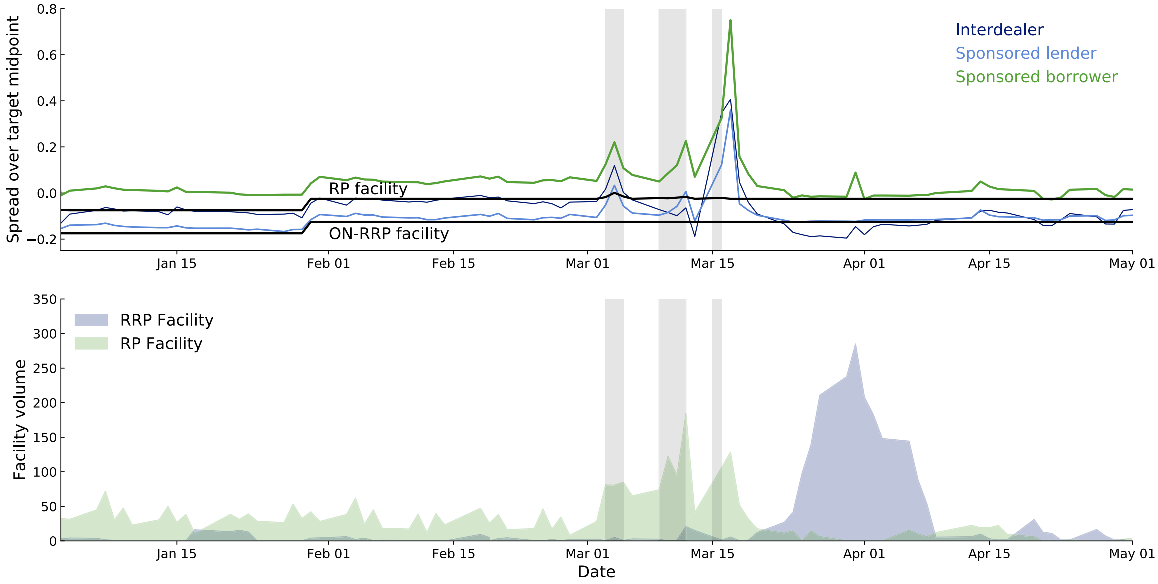

Fig. 4. DVP repo rate spreads and Federal Reserve facility participation. In the top panel, we present DVP repo rates from January to May 2020 across different segments of the market as a percentage spread over the Federal Reserve's federal funds target midpoint in percentage points. These rates have been split into three segments: the sponsored lending segment, which is largely money market funds lending to banks and dealers; the inter-member market, which is dealers and banks borrowing from and lending to each other; and, the sponsored borrowing market, which is largely hedge funds borrowing from dealers and banks. The two black lines show the rates on the Federal Reserve's repo and reverse-repo facilities, which since September 2019 the Fed has used to control rates in the instrumentation of monetary policy. In the bottom panel, we show volumes in these facilities. Repo rates are average overnight Treasury rates for each market segment, and data are from the OFR's Cleared Repo Collection. The two black lines represent the average rate offered by the Federal Reserve's Overnight Reverse-Repurchase Facility (ON-RRP) and Repo Facility (RP). In the bottom panel we present volumes in the Federal Reserve's RP and ON-RRP facilities in billions of dollars. Grey shaded areas represent days when the average rate in the RP facility was bid up beyond its minimum rate.

around 1.5 basis points on stress days. To put these numbers into perspective, during March of 2020, on the median day only 6 basis points separated the 25th and the 75th percentiles of rates in the SOFR, a broad index of overnight Treasury repo funding rates, and for the full year preceding March 2020 18 basis points was the median spread between the 1st and 99th percentiles of the SOFR. The spreads we estimate for large basis traders are therefore economically large relative to the total dispersion in repo rates.

Fig. 5 plots coefficient estimates of the repo rate on a hedge fund indicator, estimated separately for each day between February 1, 2020 and April 16, 2020, along with 95% confidence intervals. This represents an even stricter version of the specification reported in column (1) of Table 3, where hedge fund repo spreads are estimated daily. For most days, hedge funds pay a small and statistically insignificant premium. However, beginning on March 10th, hedge fund repo spreads rise substantially, and remain elevated until the end of March.

In the bottom panel of Table 3, we provide some suggestive evidence of the drivers of these spreads by examining the portion of these spreads that can be accounted for by lender effects. These estimates are comparisons of the rate for large basis traders to the rates of other counterparties borrowing from the same sponsor in the same hour, many of which are other sponsored borrowers. They therefore account for the effects of balance sheet costs and more tentatively the market power of sponsors. Despite the strong benchmark this establishes, roughly 40%–70% of the interaction term persists across these specifications, and across specifications the increase for large basis traders during crisis periods is estimated to be between one and two basis points. If this increase is interpreted as representing counterparty risk, then even a one basis point difference in rates is quite large considering that in the event of default the Treasury collateral is still in the hands of the dealers. These estimates suggest that a non-trivial amount of the increase in spreads for large basis traders during March 2020 was due to perceptions of higher risk from these traders. We find qualitatively similar results for specifications with an indicator for the largest basis traders.

These results demonstrate one important consequence of hedge funds' roles as warehouses of Treasuries through the basis trade: because hedge funds do not have direct access to repo facilities which control rates, but access them instead through members, they may be exposed to widening repo spreads during periods of stress. There are at least three possible sources of the spread between interdealer rates and hedge fund rates. First, large basis traders borrow through the sponsored market in which lenders are responsible for the failure of the borrower, which means they bear counterparty risk. If the risk of a hedge fund default increased relative to other counterparties during March 2020, spreads may have widened to compensate for this increased risk. Second, as intermediary balance sheets swelled with Treasuries during March, the cost of a default on repo with Treasury collateral could have risen, since the collateral the lender would assume in the event of a default would be more Treasuries. Finally, the small number of sponsors in this market may mean they exercise market power and in turn may demand higher spreads during periods of stress.

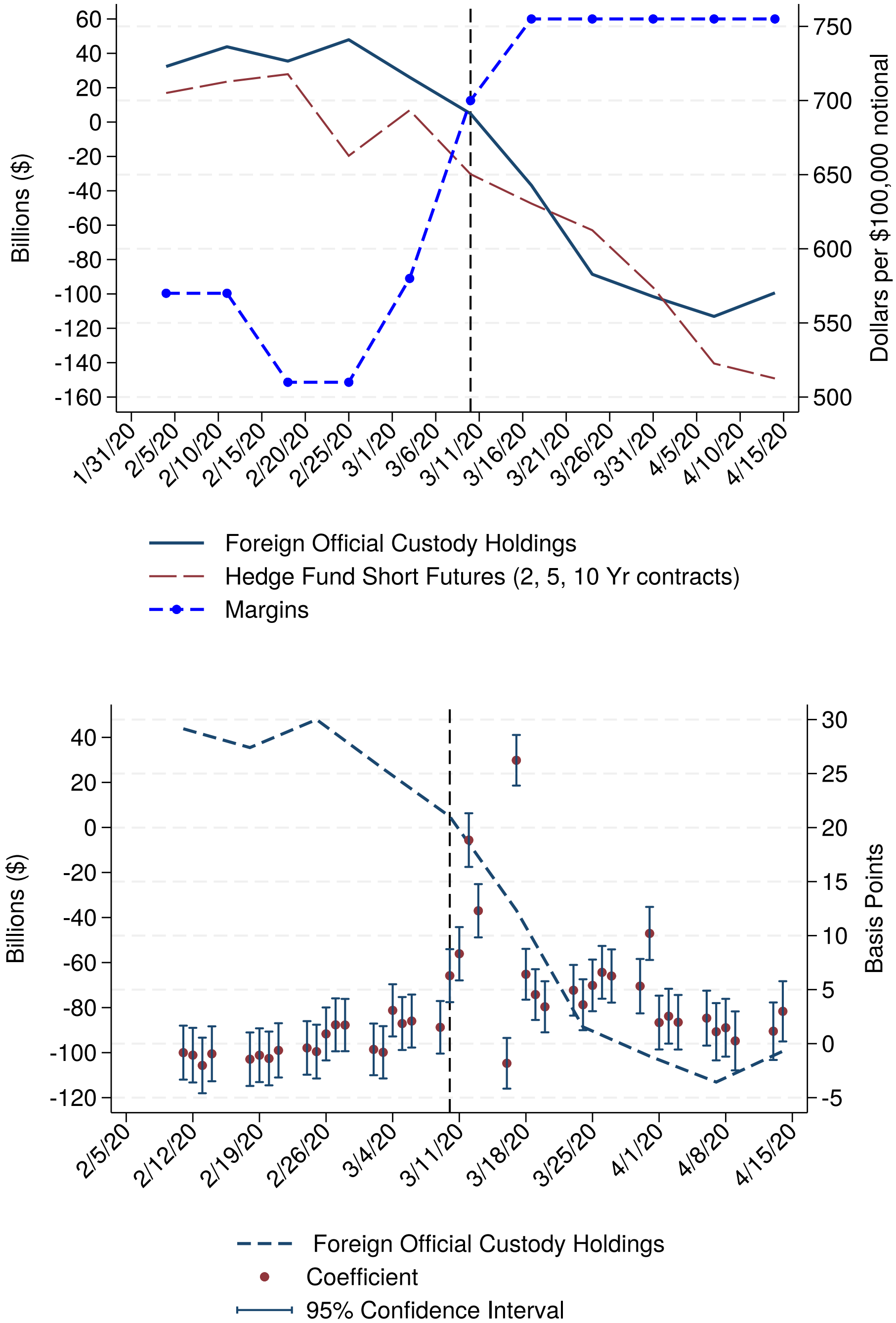

Fig. 5. Stress factors and hedge fund repo rate spreads. The top panel shows foreign central banks' holdings of Treasuries reported in billions of U.S. dollars (left axis), relative to their value in December 2019; short Treasury futures positions of hedge funds, also in billions of U.S. dollars and relative to their values in December 2019 (left axis); and maintenance margins on the 2-year Treasury futures contract per $100,000 notional amount. The bottom panel plots the same foreign official account series as in the top panel, alongside the coefficient estimate on the Large Basis Trader indicator in Eq. (3), estimated separately each day between February 1, 2020 and April 16, 2020. Coefficient estimates are akin to the panel regression estimate in column (1) of Table 3.

Table 3

DVP repo rates spreads for large basis traders. This table presents regression results for the panel of overnight Treasury repo agreements in DVP from October 2019 to May 2020. Data are aggregated to the level of collateral by lender by borrower by hour with rates being the weighted average rate for all contracts in that aggregation. Large basis trader represents a dummy for whether the borrower was classified as a large basis trader in the final quarter of 2019, and repo stress periods are periods where the repo rate was bid up above the minimum bid rate. Huber–White robust standard errors are reported in parentheses. Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

PANEL A

	Repo rates				Spread over brokered rate	
	(1)	(2)	(3)	(4)	(5)	(6)
Large basis trader	9.985***	9.606***	9.616***	8.938***	7.190***	7.068***
	(0.075)	(0.070)	(0.070)	(0.069)	(0.105)	(0.096)
Large basis trader	4.330***	2.637***	2.660***	2.575***	1.518***	1.431***
×Crisis	(0.380)	(0.481)	(0.483)	(0.484)	(0.375)	(0.330)
Observations	1,559,381	1,559,286	1,559,231	1,559,167	1,172,790	1,172,788
R-squared	0.930	0.943	0.943	0.946	0.042	0.087
Day × Hour	X	X	X	X		X
Day × Hour × OTR		X	X	X		
Day × Hour × CTD			X	X		
Day × Hour × SOMA				X		

PANEL B

	Repo rates				Spread over brokered rate
	(1)	(2)	(3)	(4)	(5)
Large basis trader	4.389***	4.453***	4.469***	4.140***	4.248***
	(0.113)	(0.098)	(0.097)	(0.095)	(0.118)
Large basis trader	1.579***	1.769***	1.808***	1.816***	0.963***
×Crisis	(0.540)	(0.621)	(0.627)	(0.632)	(0.330)
Observations	1,372,064	1,371,989	1,371,917	1,371,846	1,132,040
R-squared	0.952	0.961	0.961	0.962	0.361
Day × Hour × Lender	X	X	X	X	X
Day × Hour × OTR		X	X	X	
Day × Hour × CTD			X	X	
Day × Hour × SOMA				X	

5.5. The magnitude of the basis trade contraction

In response to increases in margins and repo rates, hedge fund short futures positions declined significantly in early March. Total hedge fund short Treasury futures declined from $749 billion to $618 billion between March 3rd and March 31, 2020, with a decline of $103 billion in the 2-year, 5-year, and 10-year short contracts.[30] A significant decline preceded March, with 2, 5, and 10-year short futures declining by $31 billion between January 28 and February 25, which may have represented some foresight of the disruptions COVID-19 could cause to Treasury markets.

Between the ends of February 2020 and March 2020, total hedge fund Treasury exposure declined from $2.38 trillion to $1.95 trillion. Estimated net sales of Treasuries were $212 billion, which we calculate as the decline in total long Treasury exposure net of long Treasury futures, minus the equivalent measure for short exposures, adjusted for rising Treasury prices.[31] For hedge funds that we classify as large basis traders, total Treasury exposure declined by $249.7 billion, from $1.56 trillion to $1.31 trillion. We are unable to estimate Treasury sales by large basis traders directly because futures data are not reported for large basis traders specifically. However, large basis trader net repo borrowing decreased by $96 billion, very similar to the $103 billion decline from 2, 5, and 10-year short Treasury futures. This suggests that nearly half of Treasury sales by hedge funds were due specifically to the unwinding of the cash-futures basis trade. These figures are provided in Table 1. Sales by basis traders were large enough that it is plausible that without timely intervention by the Federal Reserve, the unwinding of the basis trade could have further destabilized Treasury markets.[32]

The unwinding of the basis trade was likely further accelerated by the significant leverage associated with the trade. As of December 2019, large basis traders had a median leverage ratio of 15.1 and a mean leverage ratio of 16.8. At such high levels of leverage — the infamous Long Term Capital Management had leverage ratios between 25 and 30 leading into its collapse in 1998 — changes to margins or borrowing costs would pose a material threat to the equity capital of large basis traders, further motivating the trade to be unwound quickly.

[30] There are two ways to reduce futures exposure: for contracts maturing in March, hedge funds may have simply not rolled over into the June contract. For contracts maturing after March, hedge funds would need to take on offsetting long positions.

[31] To approximate the rise in Treasury prices, we calculate the percentage price change of the S&P U.S. Treasury Bond Current 10-Year Index between February 28 and March 31.

[32] We discuss this intervention in Appendix G.1.

Table 4
Cross-sectional evidence of basis trade contractions. This table reports regressions of the percentage change in long Treasury futures exposure of large basis traders between the ends of February 2020 and March 2020. The rate in sponsored is the average repo rate in the DVP market between March 9, 2020 and March 11, 2020. The change in haircuts is the percentage change in overall repo borrowing haircuts (reported in basis points), calculated as the value of total repo collateral posted divided by total repo borrowing as reported on Form PF. Unencumbered cash is the amount of unencumbered cash, scaled by net assets, of funds as of December 2019. Investor share liquidity is the weighted-average number of days investors need to redeem their shares. Huber–White robust standard errors are reported in parentheses. Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

	Dependent variable: %Δ Long treasury exposure				
	(1)	(2)	(3)	(4)	(5)
Rate in sponsored	−4.91**	−5.34**			−5.03***
	(−2.62)	(−2.83)			(−3.37)
Change in haircuts			−0.42	−0.57*	−0.49
			(−1.48)	(−1.76)	(−1.52)
Unencumbered cash		−24.02		2.86	−10.71
		(−0.84)		(0.08)	(−0.36)
Investor share liquidity		0.13*		0.15**	0.15**
		(1.94)		(2.41)	(2.45)

Even *within* large basis traders, funding market pressures appear to have mattered. In Table 4, we regress the percentage change in long Treasury exposure in the first quarter of 2020 on measures of funding costs in the FICC DVP sponsored repo segment. We include two variables of interest, the average DVP repo rate spread over the IOER between March 9 and March 11, and the percentage change in overall repo borrowing haircuts (reported in basis points), calculated as total repo collateral posted divided by total repo borrowing as reported on Form PF. We include only large basis traders, and additionally restrict to funds with non-missing values of DVP repo rates, estimated changes in aggregate haircuts between the ends of December 2019 and March 2020, and percentage changes in long Treasury exposures between December 2019 and March 2020 of between −60% and 60%. Controls include the ratio of unencumbered cash to net assets as of December 2019, a measure intended to capture funds' ability to meet sudden liquidity needs, and investor share liquidity, measured as the weighted-average number of days needed for investors to redeem capital.[33]

In column (1), we examine the average repo rate spread between March 9 and March 11, 2020. On average, a one basis-point higher repo rate spread is associated with a 4.91% larger reduction in long Treasury exposure. Including controls in column (2) increases the coefficient to 5.34%. Columns (3) and (4) look at the change in haircuts paid on Treasury collateral. In both specifications, the coefficient on haircuts is negative, suggesting a rise in haircuts is also associated with a larger reduction in long Treasury exposure, but the coefficient is only marginally significant when controls are included. In column (5), we include both repo rate spreads and haircuts along with controls. In this case, the coefficients on repo rates and changes in haircuts remain broadly unchanged, suggesting that rising repo spreads and haircuts had independent effects on hedge fund sales of Treasuries. In every specification, investor share restrictions moderate the effect of repo stress on Treasury sales, indicating the most vulnerable funds are those that faced worsening funding market pressures and were more runnable by investors.

6. Conclusion

We model how multiple markets and actors interact to determine the equilibrium size of Treasury cash-futures disconnect as well as the equilibrium quantity of arbitrage activity. We show how greater risk transfer and limits to arbitrage combine to generate a positive correlation between the cash-futures disconnect and arbitrage positions, and how these frictions may instigate an unwinding of the trade during periods of stress.

The growth of the cash-futures basis trade represents a trend of financial activity migrating to non-bank financial institutions and private capital. The greater reliance of Treasury market functioning on market-based finance may threaten the stability of Treasury markets, or other markets with substantial hedge fund arbitrage investments, during future periods of stress. Hedge funds have less regulatory oversight and more fragile capital structures than traditional intermediaries. We show that in repo markets, for instance, hedge funds face a borrowing rate penalty that grows during periods of stress, conditional on the quality of collateral. During periods of instability, hedge funds may be more likely than banks to step back from these activities or may themselves be more susceptible to forced unwinds due to deteriorating financing conditions or rising margin requirements. This paper addresses these issues through a specific arbitrage trade, but our findings speak to a broader set of issues that are likely to persist.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jmoneco.2025.103823.

[33] Form PF asks funds to report the fraction of their net assets that can be redeemed within seven time horizons: 0–1 days, 2–7 days, 8–30 days, 31–90 days, 91–180 days, 181–365 days, and more than 365 days. To construct the measure of investor share liquidity, we multiple each percentage by the respective midpoint of the range and sum.

Data availability

The data that has been used is confidential.

References

Alquist, R., Stedman, K.D., Kahn, R.J., 2022. Central Banker to the World: Foreign Reserve Management and U.S. Money Market Liquidity. Research Working Paper RWP 22-08, Federal Reserve Bank of Kansas City.
Ang, A., Gorovyy, S., van Inwegen, G.B., 2011. Hedge fund leverage. J. Financ. Econ. 102 (1), 102–106.
Aragon, G.O., Strahan, P.E., 2012. Hedge funds as liquidity providers: Evidence from the Lehman bankruptcy. J. Financ. Econ. 103 (3), 570–587.
Avalos, F., Sushko, V., 2023. Box A: Margin Leverage and Vulnerabilities in US Treasury Futures. BIS Quarterly Review International Banking and Financial Market Developments, Bank for International Settlements.
Avdjiev, S., Du, W., Koch, C., Shin, H.S., 2019. The dollar, bank leverage, and deviations from covered interest parity. Am. Econ. Rev.: Insights 1 (2).
Barth, D., Kahn, J., Monin, P., Sokolinskiy, O., 2024. Reaching for Duration and Leverage in the Treasury Market. FEDS Working Paper 2024–39, Board of Governors of the Federal Reserve System.
Boyarchenko, N., Eisenbach, T.M., Gupta, P., Shachar, O., Tassel, P.V., 2018. Bank-Intermediated Arbitrage. Staff Reports 858, Federal Reserve Bank of New York.
Boyson, N.M., Stahel, C.W., Stulz, R.M., 2010. Hedge fund contagion and liquidity shocks. J. Financ. 65, 1789–1816.
Brunnermeier, M.K., Nagel, S., 2004. Hedge funds and the technology bubble. J. Financ. 59 (5), 2013–2040.
Brunnermeier, M.K., Pedersen, L.H., 2008. Market liquidity and funding liquidity. Rev. Financ. Stud. 22 (6), 2201–2238.
Burghardt, G., Belton, T., 2005. The Treasury Bond Basis: An in-Depth Analysis for Hedgers, Speculators, and Arbitrageurs. In: McGraw-Hill Library of Investment and Finance, McGraw-Hill Education.
Chan, N., Getmansky, M., Haas, S.M., Lo, A.W., 2006. Do Hedge Funds Increase Systemic Risk? Economic Review 4, Federal Reserve Bank of Atlanta, 91.
Du, W., Tepper, A., Verdelhan, A., 2018. Deviations from covered interest rate parity. J. Financ. 73 (3).
Duffie, D., 2020. Still the World's Safe Haven? Redesigning the U.S. Treasury Market After the COVID-19 Crisis. Hutchins Center Working Paper 62, Brookings Institution.
Fleckenstein, M., Longstaff, F.A., 2020. Renting balance sheet space: Intermediary balance sheet rental costs and the valuation of derivatives. Rev. Financ. Stud. 33 (11), 5051–5091.
Garleanu, N., Pedersen, L.H., 2011. Margin-based asset pricing and deviations from the law of one price. Rev. Financ. Stud. 24 (6), 1980–2022.
Greenwood, R., Vayanos, D., 2014. Bond supply and excess bond returns. Rev. Financ. Stud. 27 (3), 663–713.
Gromb, D., Vayanos, D., 2018. The dynamics of financially constrained arbitrage. J. Financ. 73 (4), 1713–1750.
Hazelkorn, T.M., Moskowitz, T.J., Vasudevan, K., 2023. Beyond basis basics: Liquidity demand and deviations from the law of one price. J. Financ. 78 (1), 301–345.
He, Z., Nagel, S., Song, Z., 2022. Treasury inconvenience yields during the COVID-19 crisis. J. Financ. Econ. 143 (1), 57–79.
Hempel, S.J., Isley, C., Kahn, R.J., McCabe, P.E., 2023. Money Market Fund Repo and the ON RRP Facility. FEDS Notes 2023-12-15-2, Board of Governors of the Federal Reserve System.
Hempel, S., Kahn, R.J., 2021. Negative Rates in Bilateral Repo Markets. Briefs 21–03, Office of Financial Research, US Department of the Treasury.
Krishnamurthy, A., Vissing-Jorgensen, A., 2012. The aggregate demand for treasury debt. J. Political Econ. 120 (2), 233–267.
Kruttli, M.S., Monin, P.J., Petrasek, L., Watugala, S.W., 2025. LTCM redux? Hedge fund treasury trading, funding fragility, and risk constraints. J. Financ. Econ. 169, 104017.
Mitchell, M., Pedersen, L.H., Pulvino, T., 2007. Slow moving capital. Am. Econ. Rev. 97 (2), 215–220.
Nagel, S., 2016. The liquidity premium of near-money assets. Q. J. Econ. 131 (4), 1927–1971.
Pástor, L., Vorsatz, M.B., 2020. Mutual fund performance and flows during the COVID-19 crisis. Rev. Asset Pricing Stud. 10 (4), 791–833.
Peizzon, L., Subrahmanyam, M.G., Tomio, D., 2025. Central bank-driven mispricing. J. Financ. Econ. 166, 104004.
Schrimpf, A., Shin, H.S., Sushko, V., 2025. Leverage and margin spirals in fixed income markets during the Covid-19 crisis. BIS Bulletins 2.
Vissing-Jorgensen, A., 2021. The Treasury market in spring 2020 and the response of the Federal Reserve. J. Monet. Econ. 124, 19–47.
Weiss, C., 2022. Foreign Demand for U.S. Treasury Securities during the Pandemic. FEDS Notes 2022-01-28, Board of Governors of the Federal Reserve System.